

Presenter Notes

Functional Forecasting of Turkish Electricity Load Curves – v3

Slide 1 – Title

Open with the empirical object and the evaluation discipline. The thesis asks whether a transparent functional model can improve the official day-ahead benchmark. State the main result early: K=2 FPCA–VAR score model, MAE 980, 19.2% gain, pooled DM p-value near 10^{-163} .

Slide 2 – Forecasting Question

Frame the proposal as an operational benchmark problem. Positive bias means actual load exceeded the forecast. The official benchmark has average bias 467 MWh/hour over 2023–2025; the main model reduces this to 103 MWh/hour.

Slide 3 – Forecasting Object

Emphasize the modeling choice: a day is one curve, not 24 unrelated targets. Level, shape contrast, and ramp speed are properties of the full trajectory. This motivates smoothing, FPCA, score forecasting, and reconstruction.

Slide 4 – Research Design

Walk through the pipeline. Hourly observations become a smoothed function; the function is compressed into FPCA scores; scores are forecast; the curve is reconstructed. Keep the notation simple and make clear that the score forecast is the central model.

Slide 5 – FPCA Compression

The key number is 97.3% cumulative variation with two components. FPC 1 is a level factor; FPC 2 is a peak–plateau contrast. The sub-period inner products above 0.99 show that the empirical basis is stable enough for the forecasting design.

Slide 6 – Score Forecasting Model

Define FPCA–OLS carefully: FPCA scores are forecast by an OLS score equation. It is not separate hourly OLS and it is not RF. The information set is $\mathcal{F}_{d-1}$, and the model uses lag 1, lag 7, calendar, holiday, load-shape, rolling-history, and derivative features.

Slide 7 – Rolling-Origin Evaluation and DM

This slide answers whether DM can be used by hour. Yes: the pooled test uses all hourly errors, and the hour-by-hour tests apply the same loss-differential test separately for each fixed hour. Mention HAC variance and the Harvey–Leybourne–Newbold correction without over-explaining.

Slide 8 – Main Empirical Result

Read the table as a specification path. Pure score dynamics are worse than the official benchmark. Holiday and load-shape features create the large improvement; derivative features produce the main thesis model. The residual row is a second-step extension, not the baseline model.

Slide 9 – Hourly Diagnostics

The main model wins most hours but not all hours. It has lower MAE at 19 of 24 hours and DM-significant gains at 18 of 24. The weaker window is a transition period, so the correct statement is local weakness, not model failure.

Slide 10 – Residual Second Step

The point is residual structure. The base-model errors are autocorrelated, so a correction curve estimated from past residuals improves forecasts. Stress that shrinkage is selected by rolling-origin cross-validation on pre-target data only.

Slide 11 – Random Forest Benchmark

Be precise. This is the 2023 benchmark sample, not the full 2023–2025 thesis window. The FPCA comparator is the $K=2$ FPCA–VAR score model without residual correction, restricted to the same 2023 dates. Direct RF predicts each hour directly, bypasses FPCA, and is a strong nonlinear benchmark: MAE 718, gain 31.8%, lower MAE at 24/24 hours, pairwise DM vs FPCA–VAR about 1.2×10^{-85} .

Slide 12 – What Each Comparison Answers

Use this slide to prevent the RF result from confusing the thesis claim. Main FPCA–VAR vs official answers the interpretable functional model question. Residual correction answers whether the base errors contain serial structure. Direct RF answers how strong a direct nonlinear hourly benchmark is on the 2023 sample.

Slide 13 – Contributions and Status

Close the proposal candidly. The empirical pipeline is complete: FPCA representation, rolling-origin evaluation, DM testing, residual diagnostics, and RF benchmarking. Remaining work is literature positioning, final writing, and the scope decision on prediction intervals.

Slide 14 – References

Use this slide for questions. Diebold–Mariano and Harvey–Leybourne–Newbold support the test; Kokoszka–Reimherr and Hyndman–Ullah support functional forecasting; Xie–Hong is useful for the residual-correction context.